Aggregation of shift importance, and primary sources

Federated FSDS and the ten-site concept-drift benchmark

This note joins two designs that share one upload. Appendix 11 of the manuscript aggregates a simplex variable-importance vector across sites. The ten-site benchmark asks which sites carry a new concept before that aggregate is formed. Covariate shift is a change in $P(X)$. Concept drift is a change in $P(Y \mid X)$ between D_{exist} and D_{new}.

1 One-shot aggregation, then primary sources

Appendix 11 aggregates a variable-importance vector that never leaves its site. The ten-site benchmark in Section 2 asks a prior question: which of those sites carry a new concept, and which of them would corrupt the aggregate if they were treated as another noisy copy of the same shift. The two designs share the upload and the target. They do not share the failure mode.

Each site k computes a local vector $V_k \in [0,1]^p$ with $\sum_j V_{k,j} = 1$, and uploads it once. The aggregate is a convex combination

$$\bar{V} = \sum_{k=1}^{K} w_k V_k, \qquad w_k \geq 0, \qquad \sum_k w_k = 1, \tag{1}$$

or a coordinate-wise median of the same uploads. Uniform weights are $w_k = 1/K$. Sample-size weights are $w_k \propto n_k$. Inverse-variance weights use a per-site variance estimate, floored so that a vanishing variance cannot take the whole mass. The column $n|S|$ is the size-and-support weight in the simulation grid. Its gain over the uniform average is

$$\Delta(n|S| - \text{unif}) = \rho(n|S|) - \rho(\text{unif}). \tag{2}$$

Client-agreement Shapley, in the corruption table below, is a weight on sites. It is not Delta-Shapley. Delta-Shapley is the feature-wise absolute difference of Shapley values between two batches, and it appears only in the single-batch TabReD comparison. Rank aggregation averages within-site ranks. Communication is $O(Kp)$ floats. No raw row moves. Under the simplex constraint the Matrix Bernstein bound of Appendix 11 controls $\|\bar{V} - \mathbb{E}\bar{V}\|_2$ with a leading factor $(\log(2p/\delta)/K_{\text{eff}})^{1/2}$, and the coordinate-wise refinement replaces the spectral norm by an ℓ_∞ bound that is enough for a Top-K corollary when population shares are separated by a fixed margin. The bound does not choose among the mean, the median, and inverse-variance weights. The tables do.

Throughout, ρ is Spearman correlation with a stated reference. The reference changes with the table, and the change is the point. Centralized importance, the pooled out-of-distribution wrapper, and whole-batch FSDS are three different targets. A high ρ against one of them is not a high ρ against the others. Selection AUC and precision at K ask whether the top coordinates of $\bar{V}$ cover a planted support. They can stay perfect after ρ has already fallen. Pairwise concordance asks whether sites agree with each other. On several grids below it moves in the opposite direction from ρ against the pool.

Table 1: Mixed synthetic blocks with corrupted client vectors. Spearman correlation with the centralized importance. Shapley is client-agreement weight, not feature-wise Delta-Shapley.

Attack	# bad	Uniform	n-weighted	Median	Shapley
Random	1	0.208	0.539	0.798	0.802
Random	2	0.055	0.156	0.741	0.737
Extreme	1	0.503	0.576	0.830	0.814
Extreme	2	0.624	0.669	0.794	0.762
Collude	2	0.572	0.612	0.899	0.842

Table 2: FSDS on a WhyShift table, ten folds. Spearman correlation with the pooled out-of-distribution wrapper. The MMD-LOCO inverse-variance cell is not reported.

Wrapper output	Single fold	n-weighted	IVW
RF domain classifier	0.848	0.887	0.918
MMD-LOCO	0.891	0.985	—
PO-risk	0.410	0.185	0.431

1.1 Corrupted uploads

Table 1 is a weight problem, not a sample-size problem. Two random corruptions drop the uniform Spearman from a usable ranking to 0.055, and sample-size weights only to 0.156. The coordinate-wise median stays at 0.741, and client-agreement Shapley at 0.737. Extreme corruption and a two-site collusion are milder for the mean than random replacement is, and the median remains the highest or the second highest entry in every row. A corrupted vector is not a small site. Down-weighting small n_k does not remove it. A robust summary of the coordinates does.

1.2 WhyShift folds, and where PO-risk stops aggregating

Table 2 is the covariate arm and the concept arm on the same folds. A single fold of the random-forest domain classifier already correlates 0.848 with the pooled wrapper; sample-size weights move that to 0.887 and inverse-variance weights to 0.918. Leave-one-coordinate MMD moves from 0.891 to 0.985 under sample-size weights. Both wrappers are estimating a property of $P(X)$ that is shared across random folds, so averaging reduces noise. Pseudo-outcome risk does not follow them. A single fold correlates only 0.410 with the pooled PO-risk vector. Sample-size weighting lowers that correlation to 0.185. Inverse-variance weighting returns it to 0.431, essentially the single-fold number. Averaging made the concept vector worse, and a variance weight did not repair it. Section 2 is the design that explains the gap. Random folds of one concept are what the domain classifier and MMD see. A concept vector is large on a drifted site because the local regression changed, and it can also be large on a site whose labels are noise, because a flat predictor sits far from a sharp reference. Those two uploads do not estimate the same target. Inverse-variance weights assume that they do, and that the variance estimate says which copy is cleaner. Table 2 is what happens when that assumption fails.

1.3 High-dimensional corruption

The product grid crosses the number of corrupted sites n_{bad}, the local-support fraction α, and the signal-to-noise ratio on the corrupted sites. The bad-site weight w_{bad} is 0 at $n_{\text{bad}} = 0$ and 1 at

Table 3: Boundary rows of Table 4. ρ is Spearman with the planted ranking. AUC and P are selection AUC and precision at K. w_{bad} is the weight mass on corrupted sites in the inverse-variance aggregate. The variance ratio is corrupted over clean.

n_{bad}	α	SNR	ρ mean	ρ IVW	P mean	P med	w_{bad}
0	0.25	0.5	1.000	1.000	1.00	0.29	0.00
1	1	0.5	0.959	0.844	1.00	1.00	0.17
11	1	0.05	0.677	0.651	0.92	0.14	0.97
12	1	0.5	0.719	0.710	1.00	1.00	1.00
12	1	0	0.606	0.601	0.00	0.02	1.00

$n_{\text{bad}} = 12$, and the variance ratio is blank at both extremes, so those two rows are the empty-group endpoints of the grid. Read as an inverse-variance weight, $w_{\text{bad}} = 0.17$ at one corrupted site and a variance ratio of 0.44 is the share one site receives among twelve when its variance is 0.44 times the clean-site variance. Equal share would have been 1/12. Entries are means over five replicates. The full grid is Table 4. Five rows carry the reading.

Selection and rank correlation break at different places. At full support and SNR = 0.5, precision of the mean, the median, and inverse-variance weights is still 1 when every site is corrupted, while Spearman of the mean has already fallen from 1 to 0.719. The ranking gets noisier before the top set is lost. At SNR = 0 and twelve corrupted sites the selection AUC of the mean falls to 0.479 and its precision to 0. There is no remaining clean concept to recover. One clean site is enough for the mean and not for the median: at $n_{\text{bad}} = 11$, SNR = 0.05, and $\alpha = 1$, precision of the mean is 0.92 and precision of the median is 0.14. Thin support hurts the median before any site is corrupted. At $\alpha = 0.25$, SNR = 0.5, and $n_{\text{bad}} = 0$, median precision is 0.29 and median AUC is 0.768, against precision 1 and AUC 1 for the mean and for inverse-variance weights. The manuscript's statement that the median is fragile under partial overlap is this cell, not a cell with a large n_{bad}.

Inverse-variance weights pay for the variance ratio. At one corrupted site, $\alpha = 1$, and SNR = 0.5, the variance ratio is 0.44 and the bad-site weight is 0.17, about twice the equal share 1/12. Spearman falls to 0.844 against 0.959 for the uniform mean, while precision stays 1. The same pattern repeats down the SNR column: corrupted sites report a smaller variance, receive more than their count share, and the rank correlation moves before the selected set does. Flooring the variance, which the Matrix Bernstein argument already requires, is the condition under which this inflation stays bounded. It does not make inverse-variance weights a detector of a bad concept.

Table 4: Product $n_{\text{bad}} \times \text{SNR}_{\text{low}} \times \alpha \times$ aggregator on the high-dimensional stress test. Entries are means over five replicates.

n_{bad}	α	SNR	ρ mean	ρ rank	ρ med	ρ IVW	AUC mean	AUC med	AUC IVW	P mean	P med	P IVW	w_{bad}	σ^2 ratio
0	1	0.5	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—
1	1	0.5	0.959	0.977	0.962	0.844	1.000	1.000	1.000	1.00	1.00	1.00	0.17	0.44
2	1	0.5	0.925	0.955	0.928	0.772	1.000	1.000	1.000	1.00	1.00	1.00	0.31	0.43
3	1	0.5	0.894	0.935	0.891	0.730	1.000	1.000	1.000	1.00	1.00	1.00	0.43	0.43
4	1	0.5	0.864	0.912	0.857	0.709	1.000	1.000	1.000	1.00	1.00	1.00	0.53	0.44
5	1	0.5	0.836	0.889	0.823	0.692	1.000	1.000	1.000	1.00	1.00	1.00	0.61	0.45
6	1	0.5	0.812	0.866	0.795	0.689	1.000	1.000	1.000	1.00	1.00	1.00	0.69	0.45
7	1	0.5	0.792	0.842	0.759	0.689	1.000	1.000	1.000	1.00	1.00	1.00	0.76	0.44
8	1	0.5	0.776	0.822	0.732	0.691	1.000	1.000	1.000	1.00	1.00	1.00	0.82	0.44
9	1	0.5	0.761	0.800	0.702	0.699	1.000	1.000	1.000	1.00	1.00	1.00	0.87	0.43
10	1	0.5	0.745	0.779	0.674	0.698	1.000	1.000	1.000	1.00	1.00	1.00	0.92	0.42
11	1	0.5	0.734	0.761	0.655	0.709	1.000	1.000	1.000	1.00	1.00	1.00	0.97	0.38
12	1	0.5	0.719	0.737	0.631	0.710	1.000	1.000	1.000	1.00	1.00	1.00	1.00	—
0	1	0.15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—
1	1	0.15	0.938	0.973	0.957	0.768	1.000	1.000	1.000	1.00	1.00	1.00	0.18	0.39
2	1	0.15	0.890	0.946	0.911	0.691	1.000	1.000	1.000	1.00	1.00	1.00	0.33	0.39
3	1	0.15	0.852	0.920	0.867	0.657	1.000	1.000	1.000	1.00	1.00	1.00	0.45	0.39
4	1	0.15	0.818	0.894	0.828	0.636	1.000	1.000	1.000	1.00	1.00	1.00	0.55	0.40
5	1	0.15	0.791	0.870	0.788	0.630	1.000	1.000	1.000	1.00	1.00	1.00	0.63	0.40
6	1	0.15	0.767	0.845	0.751	0.635	1.000	1.000	1.000	1.00	1.00	1.00	0.71	0.40
7	1	0.15	0.746	0.818	0.708	0.637	1.000	1.000	1.000	1.00	1.00	1.00	0.77	0.40
8	1	0.15	0.730	0.794	0.672	0.646	1.000	1.000	1.000	1.00	0.97	1.00	0.83	0.40
9	1	0.15	0.713	0.767	0.641	0.652	1.000	0.999	1.000	1.00	0.91	1.00	0.88	0.39
10	1	0.15	0.699	0.742	0.609	0.654	1.000	0.995	1.000	1.00	0.86	0.99	0.93	0.39
11	1	0.15	0.686	0.716	0.582	0.664	1.000	0.987	1.000	1.00	0.77	0.92	0.97	0.35
12	1	0.15	0.675	0.692	0.562	0.668	0.988	0.971	0.988	0.75	0.64	0.77	1.00	—
0	1	0.05	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—
1	1	0.05	0.936	0.973	0.955	0.759	1.000	1.000	1.000	1.00	1.00	1.00	0.18	0.39
2	1	0.05	0.887	0.947	0.911	0.686	1.000	1.000	1.000	1.00	1.00	1.00	0.33	0.39
3	1	0.05	0.844	0.919	0.862	0.645	1.000	1.000	1.000	1.00	1.00	1.00	0.46	0.39
4	1	0.05	0.808	0.891	0.818	0.623	1.000	1.000	1.000	1.00	1.00	1.00	0.55	0.39
5	1	0.05	0.780	0.866	0.775	0.614	1.000	1.000	1.000	1.00	1.00	1.00	0.63	0.40
6	1	0.05	0.755	0.838	0.730	0.620	1.000	1.000	1.000	1.00	1.00	1.00	0.71	0.40
7	1	0.05	0.732	0.809	0.691	0.620	1.000	0.989	1.000	1.00	0.87	1.00	0.77	0.40
8	1	0.05	0.715	0.782	0.649	0.627	1.000	0.968	1.000	1.00	0.68	1.00	0.83	0.40
9	1	0.05	0.701	0.754	0.615	0.638	1.000	0.920	1.000	1.00	0.45	1.00	0.88	0.39

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n_{bad}	α	SNR	ρ mean	ρ rank	ρ med	ρ IVW	AUC mean	AUC med	AUC IVW	P mean	P med	P IVW	w_{bad}	σ^2 ratio
10	1	0.05	0.689	0.722	0.577	0.643	1.000	0.828	0.999	1.00	0.25	0.92	0.93	0.39
11	1	0.05	0.677	0.685	0.538	0.651	0.998	0.721	0.953	0.92	0.14	0.53	0.97	0.35
12	1	0.05	0.622	0.642	0.506	0.615	0.602	0.593	0.603	0.04	0.03	0.04	1.00	—
0	1	0	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—
1	1	0	0.936	0.973	0.955	0.761	1.000	1.000	1.000	1.00	1.00	1.00	0.18	0.39
2	1	0	0.886	0.946	0.910	0.681	1.000	1.000	1.000	1.00	1.00	1.00	0.33	0.39
3	1	0	0.842	0.918	0.860	0.640	1.000	1.000	1.000	1.00	1.00	1.00	0.46	0.38
4	1	0	0.806	0.890	0.819	0.619	1.000	1.000	1.000	1.00	1.00	1.00	0.55	0.39
5	1	0	0.779	0.865	0.773	0.613	1.000	1.000	1.000	1.00	1.00	1.00	0.64	0.40
6	1	0	0.754	0.837	0.730	0.619	1.000	1.000	1.000	1.00	1.00	1.00	0.71	0.40
7	1	0	0.731	0.808	0.691	0.620	1.000	0.990	1.000	1.00	0.80	1.00	0.77	0.40
8	1	0	0.713	0.778	0.647	0.627	1.000	0.950	1.000	1.00	0.58	1.00	0.83	0.40
9	1	0	0.698	0.747	0.608	0.637	1.000	0.886	1.000	1.00	0.34	0.99	0.88	0.39
10	1	0	0.689	0.711	0.567	0.645	1.000	0.763	0.999	1.00	0.17	0.93	0.93	0.39
11	1	0	0.677	0.668	0.524	0.650	0.996	0.621	0.928	0.89	0.06	0.44	0.97	0.35
12	1	0	0.606	0.624	0.491	0.601	0.479	0.478	0.478	0.00	0.02	0.00	1.00	—
0	0.75	0.5	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—
1	0.75	0.5	0.958	0.976	0.964	0.855	1.000	1.000	1.000	1.00	1.00	1.00	0.16	0.45
2	0.75	0.5	0.926	0.954	0.931	0.785	1.000	1.000	1.000	1.00	1.00	1.00	0.31	0.44
3	0.75	0.5	0.896	0.930	0.896	0.746	1.000	1.000	1.000	1.00	1.00	1.00	0.43	0.44
4	0.75	0.5	0.866	0.907	0.859	0.722	1.000	1.000	1.000	1.00	1.00	1.00	0.52	0.44
5	0.75	0.5	0.840	0.885	0.825	0.705	1.000	1.000	1.000	1.00	1.00	1.00	0.61	0.45
6	0.75	0.5	0.818	0.863	0.794	0.697	1.000	1.000	1.000	1.00	1.00	1.00	0.69	0.45
7	0.75	0.5	0.796	0.842	0.763	0.690	1.000	1.000	1.000	1.00	1.00	1.00	0.76	0.44
8	0.75	0.5	0.775	0.816	0.735	0.688	1.000	1.000	1.000	1.00	1.00	1.00	0.82	0.44
9	0.75	0.5	0.759	0.794	0.702	0.694	1.000	1.000	1.000	1.00	1.00	1.00	0.87	0.43
10	0.75	0.5	0.742	0.771	0.672	0.696	1.000	1.000	1.000	1.00	1.00	1.00	0.92	0.42
11	0.75	0.5	0.723	0.748	0.654	0.696	1.000	1.000	1.000	1.00	1.00	1.00	0.96	0.44
12	0.75	0.5	0.712	0.729	0.633	0.703	1.000	1.000	1.000	1.00	1.00	1.00	1.00	—
0	0.75	0.15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—
1	0.75	0.15	0.942	0.972	0.957	0.799	1.000	0.998	1.000	1.00	0.99	1.00	0.17	0.42
2	0.75	0.15	0.898	0.946	0.918	0.718	1.000	0.998	1.000	1.00	0.99	1.00	0.32	0.41
3	0.75	0.15	0.858	0.919	0.874	0.674	1.000	0.997	1.000	1.00	0.98	1.00	0.44	0.41
4	0.75	0.15	0.827	0.892	0.834	0.657	1.000	0.996	1.000	1.00	0.98	1.00	0.54	0.41
5	0.75	0.15	0.795	0.865	0.790	0.641	1.000	0.993	1.000	1.00	0.94	1.00	0.62	0.42
6	0.75	0.15	0.770	0.839	0.756	0.636	1.000	0.986	1.000	1.00	0.89	1.00	0.70	0.42
7	0.75	0.15	0.748	0.815	0.722	0.635	1.000	0.975	1.000	1.00	0.85	1.00	0.77	0.41
8	0.75	0.15	0.729	0.787	0.690	0.639	1.000	0.969	1.000	1.00	0.80	0.99	0.83	0.42

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n_{bad}	α	SNR	ρ mean	ρ rank	ρ med	ρ IVW	AUC mean	AUC med	AUC IVW	P mean	P med	P IVW	w_{bad}	σ^2 ratio
9	0.75	0.15	0.716	0.760	0.652	0.649	0.999	0.952	0.999	0.98	0.72	0.98	0.88	0.41
10	0.75	0.15	0.701	0.737	0.618	0.655	0.997	0.938	0.995	0.94	0.63	0.93	0.93	0.40
11	0.75	0.15	0.688	0.712	0.602	0.659	0.985	0.915	0.984	0.86	0.55	0.80	0.96	0.41
12	0.75	0.15	0.672	0.685	0.571	0.663	0.945	0.893	0.945	0.60	0.44	0.59	1.00	—
0	0.75	0.05	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—
1	0.75	0.05	0.941	0.972	0.956	0.798	1.000	0.993	1.000	1.00	0.98	1.00	0.17	0.42
2	0.75	0.05	0.895	0.945	0.913	0.712	1.000	0.992	1.000	1.00	0.99	1.00	0.32	0.41
3	0.75	0.05	0.855	0.916	0.869	0.672	1.000	0.989	1.000	1.00	0.94	1.00	0.44	0.41
4	0.75	0.05	0.821	0.890	0.830	0.650	1.000	0.981	1.000	1.00	0.89	1.00	0.54	0.41
5	0.75	0.05	0.789	0.861	0.786	0.631	1.000	0.973	1.000	1.00	0.78	1.00	0.63	0.41
6	0.75	0.05	0.767	0.836	0.749	0.633	1.000	0.943	1.000	1.00	0.72	1.00	0.70	0.41
7	0.75	0.05	0.744	0.807	0.707	0.633	1.000	0.919	1.000	1.00	0.58	0.99	0.77	0.41
8	0.75	0.05	0.723	0.776	0.674	0.634	1.000	0.885	0.999	1.00	0.43	0.96	0.83	0.41
9	0.75	0.05	0.709	0.744	0.631	0.642	0.993	0.802	0.991	0.98	0.28	0.89	0.88	0.41
10	0.75	0.05	0.693	0.713	0.593	0.646	0.968	0.723	0.958	0.92	0.13	0.77	0.93	0.40
11	0.75	0.05	0.674	0.680	0.565	0.644	0.890	0.638	0.877	0.76	0.06	0.61	0.96	0.41
12	0.75	0.05	0.630	0.639	0.524	0.622	0.589	0.527	0.589	0.04	0.01	0.05	1.00	—
0	0.75	0	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—
1	0.75	0	0.941	0.972	0.956	0.796	1.000	0.991	1.000	1.00	0.98	1.00	0.17	0.42
2	0.75	0	0.895	0.946	0.917	0.712	1.000	0.994	1.000	1.00	0.98	1.00	0.32	0.41
3	0.75	0	0.855	0.916	0.871	0.669	1.000	0.990	1.000	1.00	0.95	1.00	0.44	0.41
4	0.75	0	0.822	0.890	0.830	0.646	1.000	0.983	1.000	1.00	0.86	1.00	0.54	0.41
5	0.75	0	0.790	0.861	0.788	0.629	1.000	0.966	1.000	1.00	0.79	1.00	0.63	0.41
6	0.75	0	0.767	0.835	0.748	0.632	1.000	0.933	1.000	1.00	0.67	1.00	0.70	0.41
7	0.75	0	0.745	0.805	0.704	0.631	1.000	0.890	1.000	1.00	0.58	0.99	0.77	0.41
8	0.75	0	0.726	0.774	0.672	0.635	1.000	0.855	0.998	0.99	0.35	0.96	0.83	0.41
9	0.75	0	0.709	0.738	0.629	0.642	0.990	0.775	0.987	0.98	0.22	0.90	0.88	0.41
10	0.75	0	0.693	0.705	0.586	0.646	0.962	0.673	0.948	0.91	0.10	0.75	0.93	0.40
11	0.75	0	0.671	0.669	0.557	0.641	0.866	0.569	0.850	0.76	0.02	0.52	0.96	0.41
12	0.75	0	0.616	0.626	0.515	0.608	0.480	0.457	0.480	0.01	0.00	0.01	1.00	—
0	0.5	0.5	1.000	1.000	1.000	1.000	1.000	0.972	1.000	1.00	0.79	1.00	0.00	—
1	0.5	0.5	0.963	0.975	0.961	0.894	1.000	0.972	1.000	1.00	0.80	1.00	0.15	0.51
2	0.5	0.5	0.930	0.951	0.924	0.826	1.000	0.972	1.000	1.00	0.80	1.00	0.28	0.50
3	0.5	0.5	0.901	0.928	0.890	0.785	1.000	0.968	1.000	1.00	0.81	1.00	0.39	0.51
4	0.5	0.5	0.876	0.906	0.853	0.758	1.000	0.972	1.000	1.00	0.81	1.00	0.49	0.51
5	0.5	0.5	0.855	0.883	0.823	0.748	1.000	0.968	1.000	1.00	0.81	1.00	0.58	0.51
6	0.5	0.5	0.833	0.861	0.788	0.735	1.000	0.967	1.000	1.00	0.81	1.00	0.66	0.50
7	0.5	0.5	0.816	0.842	0.761	0.735	1.000	0.969	1.000	1.00	0.82	1.00	0.72	0.52

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n_{bad}	α	SNR	ρ mean	ρ rank	ρ med	ρ IVW	AUC mean	AUC med	AUC IVW	P mean	P med	P IVW	w_{bad}	σ^2 ratio
8	0.5	0.5	0.796	0.821	0.733	0.727	1.000	0.971	1.000	1.00	0.81	1.00	0.78	0.53
9	0.5	0.5	0.781	0.802	0.711	0.728	1.000	0.974	1.000	1.00	0.82	1.00	0.85	0.51
10	0.5	0.5	0.763	0.777	0.678	0.725	1.000	0.975	1.000	1.00	0.82	1.00	0.90	0.51
11	0.5	0.5	0.746	0.756	0.652	0.722	1.000	0.973	1.000	1.00	0.83	1.00	0.96	0.48
12	0.5	0.5	0.732	0.734	0.630	0.723	1.000	0.973	1.000	1.00	0.81	1.00	1.00	—
0	0.5	0.15	1.000	1.000	1.000	1.000	1.000	0.972	1.000	1.00	0.79	1.00	0.00	—
1	0.5	0.15	0.954	0.974	0.958	0.862	1.000	0.977	1.000	1.00	0.78	1.00	0.15	0.48
2	0.5	0.15	0.915	0.947	0.917	0.787	1.000	0.972	1.000	1.00	0.78	1.00	0.29	0.48
3	0.5	0.15	0.885	0.925	0.877	0.747	1.000	0.964	1.000	1.00	0.74	1.00	0.40	0.49
4	0.5	0.15	0.857	0.900	0.837	0.720	1.000	0.959	1.000	1.00	0.66	1.00	0.50	0.49
5	0.5	0.15	0.833	0.874	0.796	0.708	1.000	0.944	1.000	0.99	0.62	0.99	0.59	0.48
6	0.5	0.15	0.811	0.850	0.758	0.699	1.000	0.934	1.000	0.99	0.58	0.98	0.67	0.48
7	0.5	0.15	0.795	0.828	0.728	0.701	0.999	0.917	0.999	0.97	0.54	0.96	0.73	0.49
8	0.5	0.15	0.777	0.804	0.690	0.697	0.985	0.902	0.985	0.94	0.49	0.94	0.79	0.51
9	0.5	0.15	0.760	0.781	0.662	0.698	0.980	0.893	0.980	0.90	0.42	0.89	0.86	0.48
10	0.5	0.15	0.741	0.756	0.635	0.695	0.962	0.868	0.963	0.80	0.36	0.80	0.91	0.49
11	0.5	0.15	0.724	0.730	0.608	0.694	0.936	0.850	0.937	0.64	0.31	0.66	0.96	0.46
12	0.5	0.15	0.706	0.708	0.581	0.693	0.890	0.817	0.890	0.37	0.27	0.37	1.00	—
0	0.5	0.05	1.000	1.000	1.000	1.000	1.000	0.972	1.000	1.00	0.79	1.00	0.00	—
1	0.5	0.05	0.954	0.973	0.956	0.862	1.000	0.970	1.000	1.00	0.76	1.00	0.15	0.48
2	0.5	0.05	0.916	0.946	0.913	0.788	1.000	0.951	1.000	1.00	0.70	1.00	0.29	0.48
3	0.5	0.05	0.885	0.923	0.872	0.749	1.000	0.928	1.000	1.00	0.62	0.99	0.40	0.48
4	0.5	0.05	0.857	0.896	0.829	0.722	1.000	0.911	1.000	1.00	0.55	0.99	0.50	0.48
5	0.5	0.05	0.833	0.869	0.788	0.710	0.997	0.885	0.997	0.99	0.48	0.98	0.59	0.48
6	0.5	0.05	0.808	0.841	0.750	0.698	0.997	0.856	0.996	0.98	0.38	0.96	0.67	0.48
7	0.5	0.05	0.791	0.818	0.720	0.697	0.987	0.824	0.986	0.96	0.31	0.94	0.73	0.49
8	0.5	0.05	0.772	0.790	0.678	0.691	0.966	0.785	0.965	0.94	0.25	0.89	0.79	0.51
9	0.5	0.05	0.754	0.764	0.644	0.691	0.946	0.730	0.944	0.89	0.18	0.86	0.86	0.48
10	0.5	0.05	0.731	0.732	0.610	0.684	0.892	0.670	0.891	0.77	0.10	0.74	0.91	0.49
11	0.5	0.05	0.703	0.698	0.577	0.672	0.787	0.621	0.787	0.53	0.06	0.50	0.96	0.46
12	0.5	0.05	0.664	0.668	0.547	0.650	0.561	0.568	0.561	0.02	0.02	0.01	1.00	—
0	0.5	0	1.000	1.000	1.000	1.000	1.000	0.972	1.000	1.00	0.79	1.00	0.00	—
1	0.5	0	0.954	0.973	0.956	0.860	1.000	0.970	1.000	1.00	0.78	1.00	0.16	0.48
2	0.5	0	0.916	0.946	0.913	0.787	1.000	0.948	1.000	1.00	0.68	1.00	0.29	0.47
3	0.5	0	0.884	0.921	0.870	0.747	1.000	0.908	1.000	1.00	0.64	1.00	0.40	0.48
4	0.5	0	0.855	0.894	0.826	0.717	1.000	0.885	1.000	1.00	0.57	1.00	0.50	0.48
5	0.5	0	0.831	0.866	0.785	0.706	0.996	0.853	0.996	0.99	0.46	0.99	0.59	0.48
6	0.5	0	0.806	0.838	0.745	0.694	0.996	0.809	0.994	0.98	0.37	0.97	0.67	0.48

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n_{bad}	α	SNR	ρ mean	ρ rank	ρ med	ρ IVW	AUC mean	AUC med	AUC IVW	P mean	P med	P IVW	w_{bad}	σ^2 ratio
7	0.5	0	0.789	0.815	0.715	0.694	0.985	0.773	0.983	0.96	0.30	0.94	0.73	0.49
8	0.5	0	0.771	0.785	0.673	0.689	0.966	0.737	0.964	0.94	0.22	0.90	0.79	0.51
9	0.5	0	0.753	0.757	0.639	0.689	0.946	0.670	0.943	0.89	0.12	0.84	0.86	0.48
10	0.5	0	0.729	0.723	0.604	0.683	0.890	0.611	0.888	0.77	0.08	0.70	0.91	0.49
11	0.5	0	0.699	0.688	0.569	0.667	0.760	0.537	0.757	0.52	0.04	0.47	0.96	0.46
12	0.5	0	0.653	0.655	0.536	0.639	0.488	0.483	0.488	0.02	0.03	0.02	1.00	—
0	0.25	0.5	1.000	1.000	1.000	1.000	1.000	0.768	1.000	1.00	0.29	1.00	0.00	—
1	0.25	0.5	0.978	0.981	0.967	0.949	1.000	0.769	1.000	0.98	0.30	0.99	0.12	0.62
2	0.25	0.5	0.956	0.962	0.938	0.910	1.000	0.764	1.000	0.98	0.29	0.99	0.24	0.61
3	0.25	0.5	0.939	0.946	0.907	0.882	1.000	0.765	1.000	0.98	0.30	0.98	0.34	0.62
4	0.25	0.5	0.922	0.928	0.870	0.864	1.000	0.761	1.000	0.97	0.32	0.97	0.44	0.61
5	0.25	0.5	0.905	0.910	0.844	0.846	1.000	0.762	1.000	0.97	0.32	0.97	0.53	0.60
6	0.25	0.5	0.889	0.892	0.818	0.834	1.000	0.772	1.000	0.96	0.32	0.96	0.61	0.61
7	0.25	0.5	0.876	0.877	0.792	0.825	0.998	0.780	0.998	0.94	0.30	0.95	0.69	0.60
8	0.25	0.5	0.861	0.859	0.766	0.817	0.998	0.779	0.998	0.93	0.29	0.93	0.76	0.62
9	0.25	0.5	0.842	0.837	0.740	0.806	0.998	0.771	0.998	0.92	0.28	0.92	0.82	0.62
10	0.25	0.5	0.827	0.819	0.715	0.799	0.998	0.772	0.999	0.92	0.29	0.92	0.89	0.60
11	0.25	0.5	0.815	0.802	0.691	0.800	0.998	0.772	0.998	0.91	0.30	0.91	0.94	0.67
12	0.25	0.5	0.802	0.783	0.671	0.797	0.998	0.769	0.998	0.91	0.30	0.90	1.00	—
0	0.25	0.15	1.000	1.000	1.000	1.000	1.000	0.768	1.000	1.00	0.29	1.00	0.00	—
1	0.25	0.15	0.973	0.979	0.964	0.938	0.991	0.765	0.991	0.98	0.29	0.98	0.13	0.61
2	0.25	0.15	0.949	0.960	0.933	0.893	0.990	0.758	0.989	0.98	0.23	0.98	0.25	0.60
3	0.25	0.15	0.929	0.942	0.900	0.862	0.984	0.745	0.983	0.97	0.21	0.97	0.35	0.61
4	0.25	0.15	0.909	0.922	0.864	0.844	0.979	0.741	0.979	0.95	0.21	0.95	0.45	0.60
5	0.25	0.15	0.888	0.902	0.835	0.823	0.968	0.734	0.969	0.92	0.19	0.92	0.54	0.60
6	0.25	0.15	0.874	0.883	0.804	0.812	0.953	0.738	0.953	0.86	0.18	0.86	0.61	0.61
7	0.25	0.15	0.856	0.866	0.779	0.801	0.930	0.733	0.929	0.79	0.13	0.79	0.69	0.60
8	0.25	0.15	0.837	0.847	0.750	0.789	0.905	0.710	0.907	0.72	0.14	0.72	0.76	0.61
9	0.25	0.15	0.819	0.824	0.717	0.780	0.880	0.697	0.882	0.65	0.10	0.67	0.83	0.61
10	0.25	0.15	0.801	0.802	0.690	0.770	0.857	0.674	0.859	0.52	0.10	0.54	0.89	0.59
11	0.25	0.15	0.785	0.784	0.662	0.767	0.799	0.657	0.800	0.37	0.08	0.37	0.94	0.66
12	0.25	0.15	0.768	0.765	0.639	0.760	0.744	0.646	0.744	0.19	0.05	0.18	1.00	—
0	0.25	0.05	1.000	1.000	1.000	1.000	1.000	0.768	1.000	1.00	0.29	1.00	0.00	—
1	0.25	0.05	0.973	0.978	0.963	0.936	0.988	0.761	0.986	0.98	0.26	0.98	0.13	0.61
2	0.25	0.05	0.949	0.958	0.929	0.891	0.987	0.727	0.984	0.98	0.17	0.98	0.25	0.59
3	0.25	0.05	0.928	0.941	0.898	0.859	0.980	0.710	0.978	0.97	0.15	0.97	0.35	0.60
4	0.25	0.05	0.908	0.920	0.861	0.840	0.974	0.690	0.972	0.95	0.12	0.95	0.45	0.60
5	0.25	0.05	0.886	0.897	0.830	0.819	0.956	0.679	0.953	0.92	0.12	0.92	0.54	0.59

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n_{bad}	α	SNR	ρ mean	ρ rank	ρ med	ρ IVW	AUC mean	AUC med	AUC IVW	P mean	P med	P IVW	w_{bad}	σ^2 ratio
6	0.25	0.05	0.869	0.876	0.798	0.806	0.927	0.660	0.923	0.84	0.09	0.84	0.62	0.60
7	0.25	0.05	0.852	0.859	0.771	0.797	0.895	0.639	0.893	0.76	0.07	0.77	0.69	0.60
8	0.25	0.05	0.832	0.838	0.740	0.783	0.858	0.610	0.856	0.67	0.06	0.68	0.76	0.61
9	0.25	0.05	0.812	0.813	0.705	0.773	0.811	0.575	0.812	0.58	0.04	0.58	0.83	0.61
10	0.25	0.05	0.791	0.791	0.676	0.760	0.761	0.534	0.763	0.47	0.02	0.47	0.89	0.59
11	0.25	0.05	0.766	0.767	0.645	0.748	0.637	0.512	0.639	0.27	0.02	0.27	0.94	0.66
12	0.25	0.05	0.741	0.744	0.620	0.734	0.519	0.487	0.522	0.01	0.02	0.01	1.00	—
0	0.25	0	1.000	1.000	1.000	1.000	1.000	0.768	1.000	1.00	0.29	1.00	0.00	—
1	0.25	0	0.973	0.978	0.964	0.935	0.987	0.757	0.985	0.98	0.27	0.98	0.13	0.61
2	0.25	0	0.948	0.958	0.929	0.890	0.986	0.726	0.983	0.98	0.18	0.98	0.25	0.59
3	0.25	0	0.928	0.940	0.897	0.858	0.979	0.697	0.977	0.97	0.13	0.97	0.35	0.60
4	0.25	0	0.907	0.918	0.857	0.839	0.973	0.670	0.970	0.95	0.11	0.95	0.45	0.59
5	0.25	0	0.886	0.897	0.828	0.819	0.954	0.648	0.950	0.92	0.08	0.92	0.54	0.59
6	0.25	0	0.869	0.874	0.795	0.807	0.922	0.627	0.918	0.83	0.07	0.83	0.62	0.60
7	0.25	0	0.852	0.858	0.770	0.797	0.888	0.602	0.885	0.76	0.06	0.76	0.69	0.60
8	0.25	0	0.830	0.835	0.738	0.781	0.844	0.564	0.843	0.67	0.02	0.67	0.76	0.61
9	0.25	0	0.810	0.810	0.701	0.770	0.791	0.527	0.792	0.58	0.03	0.58	0.83	0.61
10	0.25	0	0.789	0.788	0.673	0.758	0.741	0.503	0.744	0.46	0.02	0.45	0.89	0.59
11	0.25	0	0.763	0.764	0.642	0.744	0.610	0.476	0.611	0.25	0.02	0.25	0.94	0.66
12	0.25	0	0.736	0.741	0.617	0.728	0.480	0.455	0.482	0.00	0.00	0.00	1.00	—

Table 5: Simulation grid (245 cells; size $\in \{0.1, \ldots, 0.9\}$, corr $\in \{0, 0.15, \ldots, 0.6\}$, hetero $\in \{0, \ldots, 1\}$): mean ρ by size-mismatch bin. Pairwise concordance is landscape only.

size bin	n	pair	unif	n-wt	$n\|S\|$	rank	$\Delta(nS - \text{unif})$
low (≤ 0.3)	70	0.241	0.651	0.660	0.655	0.658	+0.004
mid	105	0.191	0.658	0.786	0.784	0.647	+0.126
high (≥ 0.7)	70	0.105	0.620	0.905	0.905	0.580	+0.285

Table 6: Fine grid slice: mean $\rho(n|S|)$, pairwise, and $\Delta(n|S| - \text{unif})$ by size bin $\times$ hetero fraction (averaged over feature correlation). Under high size mismatch, Δ rises with hetero while pairwise falls. Alignment to the mixed pool strengthens as sites disagree.

	low size			mid			high size	
hetero	$\rho_{n\|S}$	pair	Δ	$\rho_{n\|S}$	pair	Δ	$\rho_{n\|S}$	Δ
0	0.653	0.339	+0.01	0.768	0.297	+0.11	0.893	+0.24
1/6	0.639	0.283	+0.01	0.791	0.245	+0.12	0.886	+0.25
1/3	0.656	0.262	−0.00	0.769	0.215	+0.11	0.888	+0.27
1/2	0.640	0.250	+0.00	0.773	0.197	+0.13	0.892	+0.27
2/3	0.657	0.213	−0.00	0.793	0.145	+0.13	0.921	+0.29
5/6	0.664	0.183	+0.01	0.790	0.134	+0.12	0.925	+0.31
1	0.675	0.157	+0.01	0.804	0.103	+0.15	0.927	+0.37

1.4 Size mismatch against heterogeneity

Table 5 separates two agreements. Pairwise concordance is agreement among sites. Spearman ρ is agreement with the mixed pool. In the high size-mismatch bin the sites agree least with each other (pairwise 0.105) and the size-weighted aggregates agree most with the pool (0.905, against 0.620 for the uniform average). The gain $\Delta(n|S| - \text{unif})$ is +0.004 when sizes are close and +0.285 when they are not. Rank aggregation moves the other way: 0.658 in the low bin and 0.580 in the high bin. Equal votes are the right default only when the sites are the same size.

Table 6 is the same split with the heterogeneity fraction in the rows. At low size mismatch, Δ stays inside a percentage point for every hetero value, and pairwise concordance falls from 0.339 to 0.157 as hetero goes from 0 to 1. Sites disagree more, and the pool-alignment of $n|S|$ barely changes, because the uniform average was already close. At high size mismatch the gain rises from +0.24 to +0.37 over the same hetero path, and $\rho(n|S|)$ rises from 0.893 to 0.927. Table 7 shows that this rise is not a feature correlation artifact. At hetero = 1 the gain is between +0.35 and +0.40 at every correlation from 0 to 0.60. At hetero = 0 it is between +0.19 and +0.27. Heterogeneity of the shift and disagreement among sites are not evidence that the aggregate has failed. They are the regime in which a size weight earns its keep, provided the large sites are the ones whose vector estimates the pool.

1.5 TabReD

TabReD is the balanced real-data check. The largest site fraction on the summary grid is 0.24, so the simulation says the gain of $n|S|$ over the uniform average should be small. It is.

On the nine cells the mean ρ is 0.453 uniform, 0.460 by sample size, 0.466 by $n|S|$, and 0.453 by rank. The largest single-cell gap between $n|S|$ and uniform is a few hundredths, except where

Table 7: High size-mismatch cells only: $\Delta(n|S| - \text{unif})$ on the hetero × feature-correlation grid. The size-weighting gain is large and stable across correlation; it grows further as hetero → 1.

hetero \ corr	0	0.15	0.30	0.45	0.60
0	+0.19	+0.23	+0.25	+0.27	+0.26
1/6	+0.25	+0.26	+0.27	+0.26	+0.23
1/3	+0.24	+0.27	+0.26	+0.33	+0.23
1/2	+0.22	+0.28	+0.26	+0.28	+0.31
2/3	+0.29	+0.28	+0.29	+0.31	+0.30
5/6	+0.29	+0.36	+0.30	+0.28	+0.28
1	+0.40	+0.36	+0.37	+0.35	+0.36

Table 8: TabReD federated FSDS: ρ on nine task×split cells. Site sizes are nearly balanced (max-frac ≡ 0.24), so $\Delta(n|S| - \text{unif})$ is small; $n|S|$ remains a safe default. The cooking / `num_105` cell is consensus-misaligned (high pair, negative site↔whole); see Table 9.

task	split	pair	site↔whole	unif	n	$n\|S\|$	rank	best
cooking	num_105	0.884	−0.198	−0.200	−0.191	−0.204	−0.209	n
cooking	num_80	0.383	0.161	0.250	0.305	0.329	0.251	$n\|S\|$
cooking	num_69	0.835	0.468	0.520	0.525	0.530	0.520	$n\|S\|$
ecom	num_109	0.787	0.545	0.642	0.645	0.647	0.635	$n\|S\|$
ecom	num_100	0.825	0.532	0.598	0.596	0.610	0.605	$n\|S\|$
ecom	num_103	0.851	0.470	0.498	0.505	0.516	0.501	$n\|S\|$
weather	num_38	0.598	0.428	0.554	0.514	0.531	0.546	unif
weather	num_44	0.716	0.725	0.836	0.840	0.846	0.827	$n\|S\|$
weather	num_37	0.519	0.283	0.378	0.399	0.389	0.401	med
mean	—	0.711	0.379	0.453	0.460	0.466	0.453	—

every aggregator is negative. Cooking `num_105` has pairwise concordance 0.884 and site-to-whole correlation −0.198. The sites agree with each other and disagree with the whole. Every aggregator returns a Spearman near −0.20. There is no weight that can turn a consensus of the wrong vector into the pooled vector. Table 9 locates the defect on the split. Under the feature-rank protocol, twenty replicates keep pairwise concordance 0.88 ± 0.01 and ρ at 0.07 ± 0.17 for every aggregator, with $\Delta = +0.01$. Under a random array split, two hundred replicates drop pairwise concordance to 0.26 ± 0.06 and raise ρ to about 0.80, again with Δ at 0. The headroom of the aggregator is zero on both protocols. The feature-rank axis is a shared direction that is not the whole-batch direction. Random splits destroy that shared direction and restore the usual aggregation problem, in which sites are noisy and the mean works.

The single-batch method table is not this metric. It scores each procedure against the top coefficients of a linear regression, one batch at a time.

Mean F1 against the linear top set is 0.433 for Delta-Shapley, 0.400 for PermuCATE and for LOCO, 0.367 for the generalized random forest, 0.333 for the forest, 0.311 for FSDS, and 0.178 for the knockoff. Mean selection AUC has the same lead: Delta-Shapley 0.701, forest 0.647, FSDS 0.588. The ordering is a property of one batch and one linear reference. It does not say which convex weight should merge ten uploads. The timing row is the practical counterpart. FSDS and the forest are 0.6 seconds per cell. The causal block shared by PermuCATE, LOCO, and the generalized random

Table 9: cooking-time / num_105: feature-rank versus random array-split (mean±SD of ρ). The defect is the split axis, not $n|S|$ (aggregator headroom ≈ 0).

protocol	n	pair	unif	n	$n\|S\|$	Δ
feature-rank	20	0.88 ± 0.01	0.07 ± 0.17	0.07 ± 0.17	0.07 ± 0.17	+0.01
random	200	0.26 ± 0.06	0.80 ± 0.06	0.81 ± 0.06	0.80 ± 0.06	+0.00

Table 10: TabReD method comparison: F1 vs linreg top-$|\beta|$ (PermuCATE $n_{\text{perm}} = 50$, RF outcome). Single-batch only. Not the federated metric. ΔSH is feature-wise Delta-Shapley.

task	split	FSDS	RF	ΔSH	MMD	PC	LOCO	GRF	L1_m	L1_s	KO
cooking-time	num_184	0.100	0.200	0.500	0.200	0.400	0.200	0.200	0.400	0.300	0.200
cooking-time	num_38	0.100	0.200	0.600	0.200	0.400	0.200	0.400	0.400	0.000	0.000
cooking-time	num_123	0.100	0.200	0.300	0.200	0.300	0.200	0.200	0.600	0.000	0.000
ecom-offers	num_67	0.500	0.500	0.400	0.500	0.500	0.500	0.500	0.200	0.200	0.000
ecom-offers	num_104	0.500	0.500	0.400	0.400	0.400	0.500	0.500	0.300	0.300	0.200
ecom-offers	num_60	0.400	0.500	0.200	0.400	0.400	0.500	0.400	0.300	0.400	0.100
weather	num_5	0.400	0.300	0.400	0.200	0.400	0.500	0.200	0.300	0.200	0.400
weather	num_84	0.300	0.300	0.500	0.200	0.300	0.500	0.400	0.300	0.500	0.300
weather	num_98	0.400	0.300	0.600	0.300	0.500	0.500	0.500	0.300	0.300	0.400
mean	—	0.311	0.333	0.433	0.289	0.400	0.400	0.367	0.344	0.244	0.178

forest is 56 seconds and is written three times in the total of 220.5. A joint run pays that block once.

A second TabReD partition scores the aggregate against whole-batch FSDS rather than against a linear coefficient vector. The splits are not the nine cells of Table 8.

Cooking num_184 repeats the num_105 geometry: pairwise concordance 0.875, and every aggregate correlates only about 0.12 with the whole batch. Weather num_5 and num_84 have pairwise concordance near 0.78 and a negative Spearman for every convex weight; the median is the least negative, and it is still negative. Weather num_98 is the opposite cell: pairwise concordance is the lowest on the table (0.440) and $n|S|$ reaches 0.631. Agreement among sites is not a substitute for correlation with the pool. The same sentence was the high size-mismatch row of Table 5, where pairwise concordance was 0.105 and $\rho(n|S|)$ was 0.905.

1.6 Leave-one-site-out is not a source detector

Leave-one-site-out influence answers a narrow question: if the reference vector is held fixed, how much does ρ move when one upload is deleted. On the random partition the aggregate is already aligned ($\rho_{n|S|} = 0.889 \pm 0.106$ over 45 cells) and the largest deletion effect averages 0.054. On the feature-rank partition the aggregate is not aligned ($\rho_{n|S|} = 0.243 \pm 0.241$) and the largest deletion effect is 0.057, the same magnitude. Table 15 shows the same ceiling cell by cell. Weather num_98, the best feature-rank cell ($\rho = 0.645$), moves by 0.093 under $n|S|$. Weather num_5, a negative cell, moves by 0.031. Influence does not rank the misaligned split above the aligned one. Deleting one site cannot reveal that every site is pointing the same wrong way. That is the num_105 geometry again, and it is why a source detector has to be computed from the site's own outcome model rather than from the change in a simplex average.

Table 11: TabReD: mean selection AUC by task (same run as the F1 table).

task	FSDS	RF	ΔSH	MMD	PC	LOCO	GRF	$L1_m$	$L1_s$	KO
cooking-time	0.537	0.588	0.673	0.500	0.510	0.583	0.596	0.758	0.416	0.319
ecom-offers	0.797	0.792	0.700	0.626	0.642	0.613	0.806	0.511	0.572	0.361
weather	0.430	0.561	0.729	0.588	0.682	0.687	0.629	0.512	0.510	0.536
mean	0.588	0.647	0.701	0.571	0.611	0.628	0.677	0.594	0.499	0.405

Table 12: TabReD efficiency: mean wall seconds per cell (9 cells). PermuCATE / LOCO / GRF share one causal block (time split approximately equally). The total on the last column adds that shared block three times.

	FSDS	RF	ΔSH	MMD	PC	LOCO	GRF	$L1_m$	$L1_s$	KO	total
sec	0.6	0.6	7.9	20.9	56.0	56.0	56.0	7.5	7.5	7.5	220.5
× vs first	1.0	1.0	14.2	37.6	100.9	100.9	100.9	13.5	13.5	13.5	—

1.7 What is delegated to the site score

The aggregation tables support four uses of $\bar{V}$, and they mark two places where a better weight is the wrong repair. First, when the uploads are noisy copies of one covariate shift, sample-size weights and inverse-variance weights raise Spearman with the pool, and the gain is large exactly when site sizes differ (Tables 2 and 5). Second, when a few uploads are replaced by arbitrary vectors, the median and client-agreement Shapley hold a Spearman near 0.8 after the uniform mean has fallen below 0.2 (Table 1). Third, selection of a planted support survives a large corrupted fraction whenever one clean concept remains and the local support is not thin; rank correlation moves earlier, and the median loses the top set under thin support even with no corruption (Table 3). Fourth, on balanced TabReD splits, $n|S|$ is a safe default because it ties the uniform average, not because it wins a large margin.

The two failures are not weight failures. Pseudo-outcome risk on random WhyShift folds becomes less correlated with its own pooled vector after sample-size weighting. Cooking `num_105` and `num_184` have high pairwise concordance and a near-zero or negative correlation with the whole, for every aggregator, and leave-one-site-out influence stays near 0.05 whether the aggregate is aligned or not. Both say that the sites are not noisy copies of the reference. A convex weight, a median, and a deletion diagnostic all assume that they are. The next section drops that assumption. Sites share $P(X)$ by construction, the concept is planted on five of ten sites, and one further site replaces labels by coin flips. The site score is a skill-gated excess risk against a size-matched burn-in model. It is a weight that can be computed before $\bar{V}$, and it is the object that separates a learnable concept from a consensus breaker. The aggregation results above are what that score is being asked to protect.

2 Primary-source identification across heterogeneous sites

The manuscript separates covariate shift, a change in $P(X)$, from concept drift, a change in $P(Y \mid X)$ between a reference batch D_{exist} and a new batch D_{new}. Feature-level procedures in the paper, including the pseudo-outcome risk and conditional permutation importance, score coordinates inside one batch. The federated aggregation in Appendix 11 then asks whether those variable-importance vectors can be merged without moving raw rows. That merge treats every site as a noisy look at one common shift. A prior operational question is which sites are the primary sources of the shift, and

Table 13: TabReD federated FSDS: primary metric is Spearman(agg, whole-batch), not AUC. Whole-batch means FSDS on the pooled site sample. These splits are not the cells of Table 8.

task	split	pair	ρ_{unif}	ρ_n	$\rho_{n\|S\|}$	ρ_{rank}	best
cooking-time	num_184	0.875	0.114	0.111	0.127	0.119	ns_weighted
cooking-time	num_38	0.885	0.399	0.403	0.400	0.393	n_weighted
cooking-time	num_123	0.812	0.403	0.433	0.433	0.397	ns_weighted
ecom-offers	num_67	0.769	0.321	0.362	0.356	0.300	n_weighted
ecom-offers	num_104	0.768	0.222	0.222	0.233	0.215	ns_weighted
ecom-offers	num_60	0.649	0.099	0.106	0.107	0.071	ns_weighted
weather	num_5	0.791	−0.054	−0.061	−0.057	−0.064	median
weather	num_84	0.767	−0.111	−0.118	−0.108	−0.131	median
weather	num_98	0.440	0.615	0.623	0.631	0.559	ns_weighted

Table 14: TabReD FSDS leave-one-site-out (LOSO): $\Delta_k = \rho(\text{agg}, \text{whole}) - \rho(\text{agg}_{-k}, \text{whole})$ with the whole-batch vector fixed. Report ρ and $\max_k |\Delta_k|$. Random is the deployment partition; feature-rank is the sensitivity.

protocol	task	n	$\rho_{n\|S\|}$	$\|\Delta\|_{n\|S\|}$	ρ_{unif}	$\|\Delta\|_{\text{unif}}$	$\|\Delta\|_{\text{med}}$
feature_rank	ALL	9	0.243 ± 0.241	0.057 ± 0.017	0.231 ± 0.237	0.047 ± 0.016	0.052 ± 0.029
feature_rank	cooking-time	3	0.300 ± 0.183	0.053 ± 0.008	0.299 ± 0.199	0.045 ± 0.015	0.036 ± 0.013
feature_rank	ecom-offers	3	0.264 ± 0.093	0.057 ± 0.006	0.245 ± 0.088	0.057 ± 0.005	0.061 ± 0.019
feature_rank	weather	3	0.165 ± 0.419	0.060 ± 0.031	0.148 ± 0.400	0.040 ± 0.024	0.060 ± 0.047
random	ALL	45	0.889 ± 0.106	0.054 ± 0.052	0.884 ± 0.115	0.049 ± 0.042	0.064 ± 0.048
random	cooking-time	15	0.807 ± 0.144	0.101 ± 0.068	0.794 ± 0.159	0.090 ± 0.049	0.114 ± 0.047
random	ecom-offers	15	0.921 ± 0.049	0.023 ± 0.013	0.923 ± 0.044	0.022 ± 0.012	0.030 ± 0.017
random	weather	15	0.939 ± 0.033	0.040 ± 0.014	0.934 ± 0.034	0.035 ± 0.011	0.048 ± 0.024

which sites are either stable or actively destroying the consensus. This section constructs a ten-site benchmark in which the answer is known, and evaluates a counterfactual concept score against that answer.

The design isolates concept drift. Every site is a random slice of one real covariate table, so the marginal distribution $P(X)$ is shared. The American Community Survey extracts below are cross-sections, not panels, so the batch index is a seeded random order rather than a clock. Stationarity of $P(X)$ is then a construction, and a covariate arm that fires is evidence of a bug in the generator rather than a feature of the task. Labels follow the real outcome through a burn-in and are then rewritten, on a known subset of sites, by a known rule.

2.1 Data and the ten-site generator

Two WhyShift tasks supply the base table (X, y^{base}), both taken from the 2018 ACS one-year person file for California and both using the official `folktables` feature lists. ACS Income has $p = 10$ covariates and the indicator that personal income exceeds \$50,000. ACS Public Coverage has $p = 19$ covariates and the indicator of public health-insurance coverage. In each Monte Carlo replicate the analysis sample is winsorized at the first and ninety-ninth percentiles, standardized, and cut into $K = 10$ non-overlapping sites of $n_k = 1600$ rows. Each site is then cut into $T = 16$ batches of 100 rows. The burn-in is the first four batches. The concept change, when it is switched on, begins at

Table 15: Feature-rank cells: LOSO $|\Delta|_{\max}$ by aggregator (one draw per linreg top split).

task	split	$\rho_{n\|S\|}$	$\|\Delta\|_{n\|S\|}$	$\|\Delta\|_n$	$\|\Delta\|_{\text{unif}}$	$\|\Delta\|_{\text{med}}$	$\|\Delta\|_{\text{rank}}$
cooking-time	num_184	0.092	0.054	0.047	0.061	0.027	0.031
cooking-time	num_38	0.437	0.045	0.054	0.042	0.031	0.048
cooking-time	num_123	0.373	0.060	0.062	0.032	0.051	0.053
ecom-offers	num_67	0.347	0.062	0.057	0.062	0.081	0.073
ecom-offers	num_104	0.283	0.059	0.067	0.053	0.042	0.056
ecom-offers	num_60	0.163	0.050	0.050	0.055	0.059	0.077
weather	num_5	−0.025	0.031	0.029	0.022	0.029	0.059
weather	num_84	−0.126	0.055	0.038	0.031	0.037	0.039
weather	num_98	0.645	0.093	0.068	0.067	0.114	0.054

Table 16: Rule coordinates. Z_j denotes the analysis-sample standardize of the named column. Linear sites rewrite only the upper tail of Z_1. Nonlinear sites rewrite every activated row.

Task	Role of (X_1, X_2, X_3)	X_1	X_2	X_3
ACS Income	hours, schooling, age	WKHP	SCHL	AGEP
ACS Public Coverage	age, schooling, income	AGEP	SCHL	PINCP

batch $t^\star = 8$.

The ten sites have four roles. Four clean sites keep $y = y^{\text{base}}$ at every batch. Three linear sites rewrite labels inside an upper tail of a designated coordinate X_1. Two nonlinear sites rewrite every activated row. One noise site replaces activated labels with independent Bernoulli(1/2) draws. The noise site is a consensus breaker. It is not a primary source: there is no new concept for a local model to learn. Table 16 records which coordinates carry the rules. Income uses hours worked, educational attainment, and age. Public Coverage uses age, educational attainment, and personal income. These are the continuous or ordinal columns in each official feature list. The remaining columns stay in the table as nuisance covariates.

Write Z for the standardized covariates and let $M_k(q) = \{Z_1 > Q_q(Z_1)\}$ be the upper tail at quantile q, computed on the pooled analysis sample. The main design uses $q = 0.5$. On a linear site the candidate label is the halfspace

$$y^{\text{rule}} = \mathbf{1}\{\sigma_k(\beta_{k1}Z_1 + \beta_{k2}Z_2) > c_k\} \quad \text{on } M_k(q), \tag{3}$$

and y^{base} off that tail. The three coefficient vectors are $(1, 0.35)$, $(0.25, 1)$, and $(-0.85, 0.55)$. The threshold c_k is the quantile of the linear score that matches the positive rate of y^{base} on $M_k(q)$, and the sign $\sigma_k \in \{\pm 1\}$ is the orientation whose matched labels agree less with y^{base}. The sign choice is part of the generator. It keeps the planted concept from collapsing back onto the real outcome when hours or schooling already predict income. It is not estimated by the detector.

On a nonlinear site the same rate match is applied to every row, not only the tail:

$$y^{\text{rule}} = \mathbf{1}\{Z_1 Z_2 > c\} \qquad \text{or} \qquad y^{\text{rule}} = \mathbf{1}\{\sin(Z_1) + Z_3^2 > c\}. \tag{4}$$

The first site uses the product and the second uses the sine-square rule. The planted feature support is $\{X_1, X_2\}$ for every linear site and for the product site, and $\{X_1, X_3\}$ for the sine-square site.

Activation is a batch weight $w_t \in [0, 1]$. Under abrupt activation, $w_t = \mathbf{1}\{t \geq t^\star\}$. Under gradual activation, $w_t = 0$ before $t^\star$ and then rises linearly to one over four batches. Each post-burn-in row

Table 17: Realized label changes after $t^\star$, averaged over four seeds at $\alpha = 1$, abrupt activation, and $q = 0.5$. Flip rate is the fraction of post-burn-in rows whose label differs from y^{base}. Agreement is the fraction of the rewritten region on which the rate-matched rule reproduces y^{base}. Clean sites are not rewritten, so agreement is one. The noise site has no rule, so agreement is left blank.

Task	Site type	Flip rate	Rule agreement	Post positive rate	Base positive rate
Income	Clean	0.000	1.000	0.406	0.406
Income	Linear	0.131	0.427	0.400	0.414
Income	Nonlinear	0.555	0.436	0.396	0.420
Income	Noise	0.498	—	0.491	0.410
Public Coverage	Clean	0.000	1.000	0.369	0.367
Public Coverage	Linear	0.259	0.476	0.378	0.372
Public Coverage	Nonlinear	0.492	0.506	0.371	0.374
Public Coverage	Noise	0.496	—	0.495	0.362

is rewritten with probability αw_t, and otherwise keeps y^{base}. The main grid uses $\alpha = 1$. The noise site uses the same weights and then draws a fresh coin flip. Clean sites ignore the weight.

Two consequences are worth stating before any score is computed. First, because the structured rules match the local positive rate, a marginal label discrepancy is not a substitute for concept drift. The largest average gap between the post-change positive rate and the baseline, among structured sites, is 2.4 percentage points (Income, nonlinear). Second, hours worked in the ACS piles up at 40. The empirical quantiles of WKHP at 0.5 and at 0.7 are both 40, so $\{Z_1 > Q_{0.5}\}$ and $\{Z_1 > Q_{0.7}\}$ are the same set, about 23% of Income rows. The median tail in the main Income design is therefore thinner than a continuous coordinate would suggest. Public Coverage uses age as X_1, and that tail is about half the sample. Table 17 records the realized flip rates. Structured sites move roughly half the labels inside the region they rewrite, and they leave the marginal positive rate near the baseline. The noise site flips about half of all post-burn-in labels and shifts the positive rate to one half.

2.2 The counterfactual concept score

Let $\mathcal{D}_0$ be the pooled burn-in, eight thousand rows in the main design, still generated from y^{base} at every site. Let $\mathcal{D}_k$ be the most recent six batches at site k, six hundred rows, all of them inside the drift window when activation is abrupt. A random forest with 25 trees, depth 8, and minimum leaf size 5 is the outcome model throughout, in line with the recommendation that the forest be used for distance estimation rather than for a causal reading of the meta-learner. Local predictions on $\mathcal{D}_k$ are three-fold cross-fit probabilities μ_k^{oof}. The reference probabilities μ_0^{sm} are an average of two forests, each trained on a random subset of $\mathcal{D}_0$ whose size equals the local training set, two thirds of the current window. Matching the training size stops the larger reference sample from looking like a concept change. A third forest trained on all of $\mathcal{D}_0$ is retained only as the pooled excess defined below. It is not the headline score.

On a region $\mathcal{R}$ that has both classes in the current window and in the reference, at least fifty current rows, and at least eighty reference rows, write $B(y, p)$ for the Brier score and

$$\Delta_k(\mathcal{R}) = \Big(B(\mu_0^{\text{sm}}) - B(\mu_k^{\text{oof}})\Big)_+ \cdot \Big(2\big(\text{AUC}(\mu_k^{\text{oof}}) - \tfrac{1}{2}\big)\Big)_+. \tag{5}$$

The first factor is the excess risk of the reference concept over a local concept of the same capacity. The second factor is zero when the local model does not beat a coin, and it equals one when the local

Table 18: Mean score by site type at $\alpha = 1$, abrupt activation, and $q = 0.5$. Four seeds. Δ^{cf} is (6). Δ(all) is the whole-window score. Δ(high) is the upper tail of X_1. Domain AUC is the logistic classifier on X only.

Task	Type	Δ^{cf}	Δ(all)	Δ(high)	τ^2	Skill	Domain AUC
Income	Clean	0.003	0.000	0.002	0.015	0.696	0.498
Income	Linear	0.253	0.024	0.253	0.181	0.937	0.482
Income	Nonlinear	0.336	0.298	0.316	0.313	0.897	0.494
Income	Noise	0.001	0.000	0.000	0.096	0.008	0.494
Public Coverage	Clean	0.001	0.000	0.001	0.014	0.364	0.503
Public Coverage	Linear	0.200	0.052	0.200	0.132	0.943	0.493
Public Coverage	Nonlinear	0.236	0.182	0.179	0.159	0.970	0.478
Public Coverage	Noise	0.003	0.002	0.003	0.046	0.089	0.528

AUC equals one. The headline score is the maximum over a dictionary that does not use labels,

$$\Delta_k^{\mathrm{cf}} = \max_{\mathcal{R} \in \{\text{all, high, low}\}} \Delta_k(\mathcal{R}), \tag{6}$$

where "high" and "low" are the two sides of the burn-in median of X_1. The median is computed on $\mathcal{D}_0$ only. The whole-window score Δ_k(all) is reported beside the maximum so that the cost of ignoring the tail is visible.

The same two probability vectors define the concept contrast

$$\tau_k^2 = \frac{1}{|\mathcal{D}_k \cap \mathcal{R}_k^\star|} \sum_{i \in \mathcal{D}_k \cap \mathcal{R}_k^\star} \big(\mu_k^{\mathrm{oof}}(X_i) - \mu_0^{\mathrm{sm}}(X_i)\big)^2, \tag{7}$$

with $\mathcal{R}_k^\star$ the region attaining the maximum in (6). This is the concept contrast that the pseudo-outcome decomposition isolates from the covariate arm once $P(X)$ is shared across D_{exist} and D_{new}. The benchmark does not re-estimate a propensity: under the common covariate table that weight is not where the signal sits. Gating by the same skill factor as in (5) produces $\tau_k^{2,\mathrm{g}}$. Raw τ_k^2 is retained because a flat local predictor on a noise site can still sit far from a sharp reference predictor. The gate is the part of Δ_k^{cf} that refuses to call that gap a primary source.

For a score s and two disjoint collections of sites $\mathcal{A}$ and $\mathcal{B}$, the pairwise concordance is

$$C(s; \mathcal{A}, \mathcal{B}) = \frac{1}{|\mathcal{A}|\,|\mathcal{B}|} \sum_{a \in \mathcal{A}} \sum_{b \in \mathcal{B}} \mathbf{1}\{s_a > s_b\}. \tag{8}$$

Ties count as failures. The primary set $\mathcal{P}$ is the five structured sites. The target for the headline score is $C(\Delta^{\mathrm{cf}}; \mathcal{P}, \mathcal{C}) > 0.8$, where $\mathcal{C}$ is the set of four clean sites. The same functional is reported for $\mathcal{P}$ against the noise site, and separately for the linear and nonlinear subsets of $\mathcal{P}$. Every entry below is a mean over four seeds, with the worst seed in parentheses, unless a row is explicitly a per-type mean.

Negative controls use the same windows and never see the label rule. The domain AUC is the cross-fit AUC of a classifier trained to separate burn-in rows from recent rows using X alone, once with the forest and once with logistic regression. The covariate discrepancy is an RBF MMD on subsamples of size 250, with bandwidth equal to the median of the squared Euclidean distances. A joint classifier two-sample test appends y to X before the same forest AUC is computed. Marginal label total variation is the absolute difference of positive rates between the recent window and the burn-in.

Table 19: Pairwise concordance (8) at $\alpha = 1$, abrupt activation, and $q = 0.5$. Each cell is the mean over four seeds, with the worst seed in parentheses. $\mathcal{P}$ is the five structured sites and $\mathcal{C}$ is the four clean sites.

Task	Score	$\mathcal{P} > \mathcal{C}$	$\mathcal{P} >$ noise	Linear $> \mathcal{C}$	Nonlinear $> \mathcal{C}$
Income	Δ^{cf}	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Income	Δ(all)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Income	$\tau^{2,\mathrm{g}}$	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Income	Pareto rank	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Income	τ^2	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Income	Pooled excess	1.000 (1.000)	0.400 (0.400)	1.000 (1.000)	1.000 (1.000)
Income	Label TV	0.500 (0.100)	0.050 (0.000)	0.500 (0.083)	0.500 (0.125)
Income	Joint C2ST	0.850 (0.800)	0.350 (0.200)	0.750 (0.667)	1.000 (1.000)
Income	RF domain AUC	0.487 (0.400)	0.500 (0.000)	0.521 (0.333)	0.438 (0.250)
Income	RBF MMD	0.412 (0.300)	0.500 (0.000)	0.375 (0.167)	0.469 (0.125)
Public Coverage	Δ^{cf}	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Public Coverage	Δ(all)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Public Coverage	$\tau^{2,\mathrm{g}}$	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Public Coverage	Pareto rank	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Public Coverage	τ^2	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Public Coverage	Pooled excess	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Public Coverage	Label TV	0.613 (0.350)	0.000 (0.000)	0.604 (0.417)	0.625 (0.250)
Public Coverage	Joint C2ST	0.950 (0.850)	0.450 (0.200)	0.917 (0.750)	1.000 (1.000)
Public Coverage	RF domain AUC	0.438 (0.350)	0.200 (0.000)	0.396 (0.167)	0.500 (0.250)
Public Coverage	RBF MMD	0.600 (0.500)	0.200 (0.000)	0.562 (0.417)	0.656 (0.375)

Table 18 is the raw material of the concordance claim. On both tasks, and for every linear site, the maximum is exactly the upper-tail score: $\Delta^{\mathrm{cf}} = \Delta(\mathrm{high})$. On Income that tail score is 0.253 against a whole-window score of 0.024. The whole window is mostly unrewritten rows, and a single Brier average dilutes a real halfspace by about an order of magnitude. The noise skill is 0.008 on Income and 0.089 on Public Coverage, so the gate shuts even though the ungated contrast τ^2 on Income noise (0.096) exceeds the clean contrast (0.015). Logistic domain AUCs sit on 1/2 for every type. The largest is 0.528, on the Public Coverage noise site. The forest domain AUC, not shown column by column, has type means between 0.490 and 0.514, and equals 0.514 on that same noise site. Neither is evidence of a covariate shift.

2.3 Concordance

Table 19 is the primary result. On both tasks and in every seed, $C(\Delta^{\mathrm{cf}}; \mathcal{P}, \mathcal{C}) = 1$, and the same equality holds for linear sites alone, for nonlinear sites alone, and for $\mathcal{P}$ against the noise site. The whole-window score and the skill-gated contrast $\tau^{2,\mathrm{g}}$ match this ranking at full intensity. Figure 1 plots the clean-site comparison. The covariate rows do not. Random-forest domain AUC and RBF MMD have concordances between 0.38 and 0.66, with worst seeds as low as 0.17. Label total variation is consistent with a matched positive rate: its concordance against clean sites is 0.50 on Income and 0.61 on Public Coverage, and its concordance against the noise site is 0.05 and 0.00. The noise site is the unique marginal shift, and a marginal functional ranks it first.

The joint classifier two-sample test sees any change in (X, Y). It separates $\mathcal{P}$ from $\mathcal{C}$ at 0.85 on Income and 0.95 on Public Coverage, which clears 0.8 on average, but it ranks the noise site as the stronger departure: concordance of $\mathcal{P}$ against noise is 0.35 and 0.45. On Income the linear-only comparison for the joint test falls to 0.75. A test of the joint distribution is not a test for a learnable

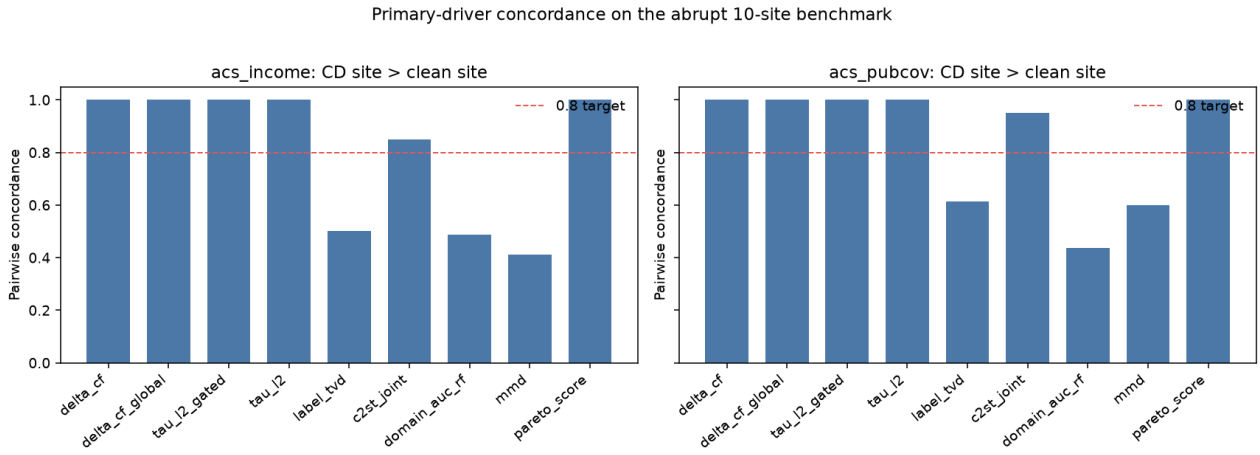


Figure 1: Pairwise concordance against clean sites. The dashed line is the 0.8 target. Δ^{cf}, the whole-window score, skill-gated τ^2, raw τ^2, and the Pareto rank clear the line on both tasks. The joint classifier two-sample test also clears it on average; its failure is the noise contrast in Table 19. Label total variation, the random-forest domain AUC, and RBF MMD do not clear the line. Source: abrupt activation, $\alpha = 1$, $q = 0.5$, four seeds.

new concept.

Pooled excess makes the same mistake in a sharper way, and only on the task where the reference concept is easy. It is the Brier gap between the forest trained on the entire burn-in and the local cross-fit model, before size matching and before the skill gate. On Income its concordance against the noise site is 0.400 in every seed. That fraction is exact: both nonlinear sites beat the noise site, and none of the three linear sites do. The pooled Income forest is a sharp predictor of y^{base}. Evaluated on coin-flip labels its Brier rises by 0.084 on average, while the same forest evaluated on a linear tail that covers 23% of the window rises by only 0.033. Size matching does not by itself fix the comparison. On the region that attains Δ^{cf}, the size-matched excess on Income noise remains 0.071, against 0.002 on clean sites. Multiplying by the skill factor is what sends the noise score from 0.071 to 0.001. Public Coverage does not reproduce the 0.40 figure. There the baseline concept is harder, the pooled excess on noise is 0.035, and every structured site still outranks it. The failure mode is real on the task where the reference fit is strong, which is the task on which an ungated risk gap is most tempting.

The Pareto comparison uses three axes when the time curve below is available, and two otherwise. The axes are Δ^{cf}, $\tau^{2,\mathrm{g}}$, and the time jump, defined as the maximum of the curve on endpoints at or after $t^\star$ minus the maximum on endpoints before $t^\star$. A site's rank is one plus the number of sites that are at least as large on every axis and strictly larger on at least one. Concordance is computed on the negative rank, so a smaller rank is a higher score. Mean ranks on Income are 1.875 for nonlinear sites, 3.250 for linear sites, 6.625 for clean sites, and 6.750 for the noise site. On Public Coverage they are 1.375, 2.667, 7.188, and 6.500. The undominated front is mostly the nonlinear sites: half of them on Income and five eighths on Public Coverage. No clean site and no noise site is undominated. Linear sites are often dominated by a nonlinear site, because the nonlinear rewrite covers every row and the linear rewrite covers a tail, but they remain strictly ahead of every clean site. That is the sense in which the Pareto order puts primary sources first. It does not claim that every primary source lies on the front.

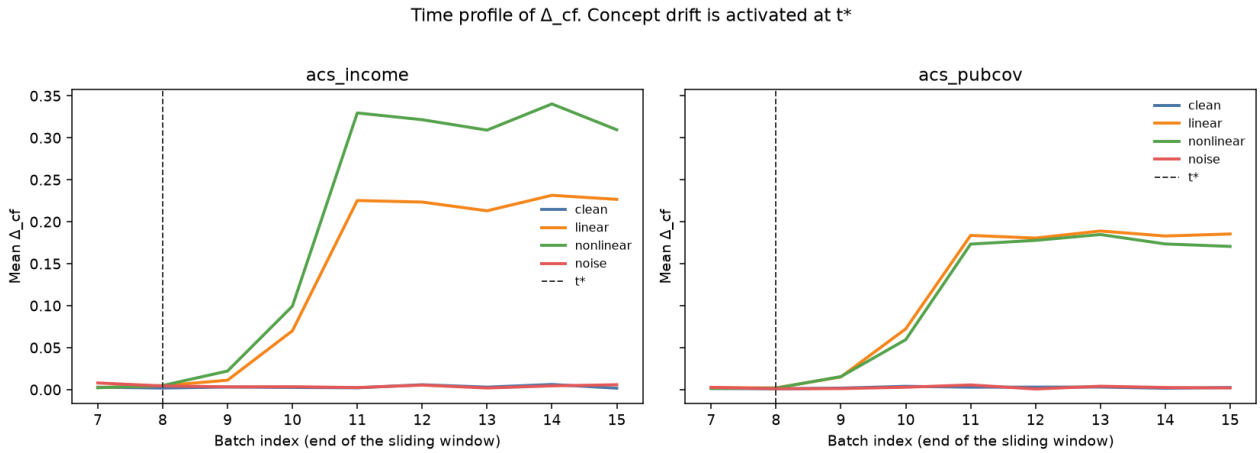


Figure 2: Mean Δ^{cf} against the last batch in a window of length four. The vertical line is $t^\star = 8$. The curve levels off at batch 11, the first endpoint whose window lies entirely after the change. Clean and noise traces remain near zero. Source: abrupt activation, $\alpha = 1$, $q = 0.5$, four seeds.

2.4 Time localization

The curve at site k reuses the burn-in reference and recomputes (6) on a sliding window of four batches, using only the whole window and the upper tail. The horizontal coordinate in Figure 2 is the last batch included in the window. Drift begins at batch 8. A window of length four is entirely inside the drift only when its last batch is at least 11. The figure follows that arithmetic. Means for clean sites and for the noise site stay below 0.01 on both sides of $t^\star$. Linear and nonlinear means leave zero between batches 9 and 10 and level off at batch 11. On Income the nonlinear plateau is near 0.32 and the linear plateau is near 0.22. On Public Coverage the two plateaus sit together near 0.18, which is consistent with the similar headline scores in Table 18.

Two detectors are applied to each curve without being told $t^\star$. The online rule calibrates on the first two evaluated endpoints, sets the threshold at that maximum plus 0.01, and reports the first later endpoint that exceeds the threshold. A hit is an estimate within two batches of $t^\star$. The derivative rule reports the endpoint of the largest first difference. Table 20 shows that the derivative is estimating the onset of the plateau rather than the onset of the drift. Its mean absolute error is about three batches on every structured type, which is one window length. The online rule is the onset estimator. It hits every nonlinear site on both tasks and every linear site on Public Coverage. Income linear sites, whose tail is the thin one, hit in 0.833 of replicates, with mean absolute error 1.92 batches. The cost is a false alarm rate of 0.375 on Income clean sites and 0.125 on Public Coverage clean sites: any crossing counts, whether or not it falls near $t^\star$. Among the Income clean curves that do cross, the mean absolute error is 4.17 batches, so the typical alarm is not a call at the change. A threshold of 0.01 above a two-point calibration is enough to rank sites when the window is the full recent block, and it is not enough to call a changepoint on a short window without a second screen. The noise site crosses the threshold in two of four Income replicates and in one of four Public Coverage replicates. One Income crossing lands on batch 10, inside the two-batch window; the other is six batches later. The Public Coverage crossing is three batches after $t^\star$ and is not counted as a hit.

Table 20: Time localization over four seeds. Online hit rate is the fraction of sites whose first threshold crossing lies within two batches of $t^\star$. False-alarm rate is computed only for clean sites. Derivative MAE is the absolute error of the largest first difference.

Task	Type	Online hit	False alarm	Online MAE	Derivative MAE
Income	Clean	0.062	0.375	4.17	4.06
Income	Linear	0.833	—	1.92	2.83
Income	Nonlinear	1.000	—	1.62	3.00
Income	Noise	0.250	—	4.00	3.25
Public Coverage	Clean	0.062	0.125	3.00	3.88
Public Coverage	Linear	1.000	—	1.42	3.00
Public Coverage	Nonlinear	1.000	—	1.50	3.00
Public Coverage	Noise	0.000	—	3.00	4.00

Table 21: Type resolution at $\alpha = 1$, abrupt activation, and $q = 0.5$. Accuracy is the fraction of structured sites whose maximizing concept group equals the planted mechanism, averaged over four seeds. Recall is pooled over both tasks and all four seeds, hence over 24 linear sites and 8 sites of each nonlinear mechanism. Top-five precision of Δ^{cf} is 1 in every seed on both tasks.

Call	Mechanism or task	Accuracy or recall
All rows	Income, accuracy	0.700
All rows	Public Coverage, accuracy	0.700
Upper tail	Income, accuracy	0.750
Upper tail	Public Coverage, accuracy	0.900
All rows	Linear, recall	0.500
All rows	Product, recall	1.000
All rows	Sine-square, recall	1.000
Upper tail	Linear, recall	0.917
Upper tail	Product, recall	0.375
Upper tail	Sine-square, recall	1.000

2.5 Concept-group type resolution

Detecting a primary source does not say whether the new concept is the linear halfspace or one of the two nonlinear rules. The dictionary is fixed before looking at site labels. On a window with outcome y, three logistic regressions are cross-fit against an intercept-only Brier score: (Z_1, Z_2), the scalar $Z_1 Z_2$, and $(\sin Z_1,\ Z_3^2)$. Group importance is the drop in Brier. The call is the maximizing group. One call uses every row in the recent window. The other uses only the upper tail of X_1, the region in which the linear rule was planted. Accuracy is computed on the five structured sites only. Top-five precision asks a different question: of the five sites with the largest Δ^{cf}, how many lie in $\mathcal{P}$.

Table 21 separates detection from typing. Top-five precision is 1 in every seed: the five largest values of Δ^{cf} are exactly the five structured sites. Conditional on that detection, the full-window dictionary recovers both nonlinear mechanisms in every replicate and recovers the linear halfspace in half of the linear sites. The miss is structural. Outside the tail the linear site still carries y^{base}, and on Income that baseline outcome is already monotone in hours and schooling. An additive logistic fit can look competitive with the product and sine-square groups even when the rewritten slice is a halfspace in (Z_1, Z_2). Restricting the call to the upper tail raises linear recall from 0.500 to 0.917

Table 22: Mean AUC of a feature score against the planted support. Linear and product supports are $\{X_1, X_2\}$. The sine-square support is $\{X_1, X_3\}$. Averages run over the structured sites of four seeds.

Task	Type	Shift permutation	Predictive permutation	ℓ_1 coefficient
Income	Linear	0.906	0.927	0.786
Income	Nonlinear	0.789	1.000	0.930
Public Coverage	Linear	0.953	0.995	0.841
Public Coverage	Nonlinear	0.985	1.000	0.879

and raises Public Coverage accuracy from 0.700 to 0.900. The same restriction lowers product recall from 1 to 0.375. The product rule was planted on every row. Inside the upper tail the factors are truncated, and the additive group in (Z_1, Z_2) is no longer dominated by the product scalar. The sine-square group remains separated on either window. Type resolution is therefore a property of the region on which the group is scored, not a single accuracy number.

2.6 Feature support and one-shot aggregation

Feature scores are computed on the recent window of each structured site and compared with that site's planted support by AUC. Four scores are used. Shift permutation importance is the drop in the squared concept contrast when a coordinate is permuted inside five quantile bins of the covariate it is most correlated with. Both the local forest and the reference forest are re-evaluated on that same permuted matrix. The local forest here is fit on the full recent window; the cross-fit in (6) is not reused. Predictive permutation importance is the drop in held-out AUC for predicting the recent label. The difference of that importance between the recent window and the burn-in is the fourth score's companion; the fourth score itself is the absolute coefficient vector of an ℓ_1-penalized logistic regression on the recent window, with $C = 0.5$.

Table 22 says that the planted coordinates are recoverable from a single drifted site. Predictive permutation AUC is at least 0.927. Shift permutation AUC is at least 0.906 on linear sites and falls to 0.789 on Income nonlinear sites, where the product and the sine-square rule are functions of coordinates that also predict the baseline income label. The ℓ_1 coefficient is weaker than the forest on linear Public Coverage (0.841) and stronger than shift permutation on Income nonlinear sites (0.930). None of these AUCs is a site-ranking result. They are the contents of the vector a site would upload.

Aggregation treats each upload as a point on the simplex. Negative shift-permutation coordinates are truncated at zero and the vector is renormalized. The target vector puts mass on a coordinate proportional to the number of structured sites that plant it. Uniform averaging, the coordinate-wise median, inverse-variance weights with variance estimated from the permutation repeats and floored at 10^{-4}, and weights proportional to Δ^{cf} are compared by AUC, precision at the number of positive target coordinates, and Spearman correlation with the target. Leaving the noise site out of the uniform average, and averaging only the five structured sites, are oracles used as ceilings rather than as methods.

On Public Coverage every shift aggregator has AUC at least 0.984, and the Δ^{cf}-weighted average and the average that drops the noise site both have AUC 1 and precision 1. On Income the uniform shift average has AUC 0.869 and precision 0.667. Weighting by Δ^{cf} raises those figures to 0.917 and 0.750. The predictive permutation upload and the ℓ_1 upload, averaged uniformly, reach AUC 0.988 on Income. That is not evidence against the concept score. On Income the planted coordinates are hours, schooling, and age, which are also the coordinates that predict y^{base} on a clean site. A clean

Table 23: Income concordance $C(\Delta^{\mathrm{cf}}; \mathcal{P}, \mathcal{C})$ under weaker rewrites and thinner linear tails. Three seeds. The worst seed is in parentheses. Quantile 0.5 is the main tail, about 23% of rows because of the mass at 40 hours.

	$q = 0.5$		$q = 0.8$		$q = 0.9$	
α	Abrupt	Gradual	Abrupt	Gradual	Abrupt	Gradual
1.0	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	0.850 (0.750)	0.867 (0.800)
0.7	1.000 (1.000)	1.000 (1.000)	0.967 (0.950)	0.950 (0.900)	0.817 (0.750)	0.683 (0.500)
0.4	0.783 (0.650)	0.750 (0.650)	0.617 (0.450)	0.600 (0.450)	0.550 (0.450)	0.550 (0.450)

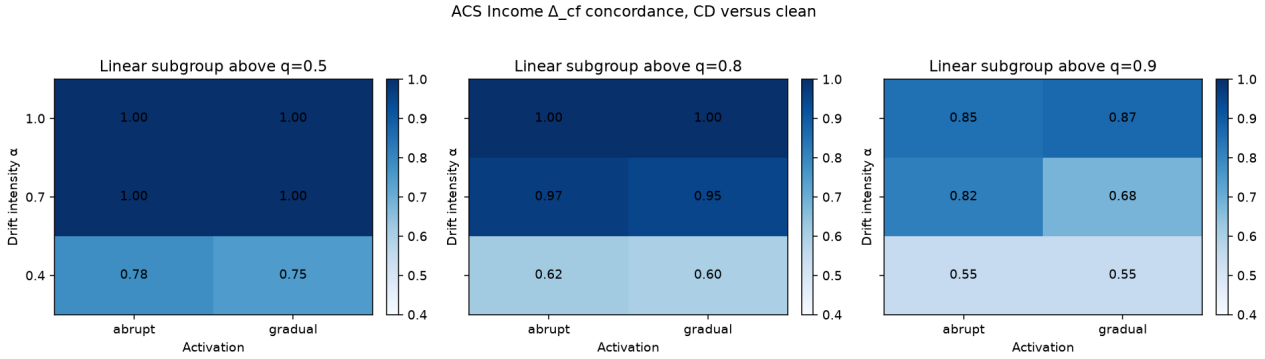


Figure 3: Income concordance of Δ^{cf} against clean sites. Each panel is one linear-tail quantile. Cell entries are means over three seeds. Means stay at or above 0.8 for $\alpha \in \{0.7, 1\}$ down to a linear tail of about 17% of rows. They fall below 0.8 when $\alpha = 0.4$, and when a tail of about 8% is combined with $\alpha = 0.7$ and a gradual ramp. Full intensity on that thin tail stays above 0.8 on average and not in every seed; the worst seeds are in Table 23.

site's predictive vector already points at the target. The shift vector is the one that is trying to upload the change rather than the baseline regression, and it is the vector that improves when clean and noise sites are down-weighted by Δ^{cf}.

Leave-one-site-out influence, defined as one minus the Spearman correlation between the uniform aggregate and the aggregate with one site removed, does not identify primary sources. Mean influence on Income is 0.033 for clean sites, 0.013 for linear sites, 0.018 for the noise site, and 0.021 for nonlinear sites. On Public Coverage every type mean lies between 0.005 and 0.008. The uploaded vectors agree with each other closely enough that deleting one site barely moves the ranking. Site-level Δ^{cf} is the functional that separates primary sources from clean sites. Influence on a simplex average is not.

2.7 Intensity, tail mass, and gradual activation

The main grid holds $\alpha = 1$, abrupt activation, and $q = 0.5$. The sensitivity grid varies those three levers on Income only, with three seeds and the same score, and it does not recompute feature uploads or the joint two-sample test. Because the 0.5 and 0.7 quantiles of hours worked are the same mass point at 40 hours, $q = 0.7$ selects the same tail as $q = 0.5$ and is omitted. The quantile 0.8 moves the threshold to 45 hours and leaves about 17% of rows in the linear tail. The quantile 0.9 moves it to 50 hours and leaves about 8%. Nonlinear sites continue to rewrite every activated row. Only the linear sites become thinner.

Table 23 and Figure 3 locate the boundary of the 0.8 target. At full intensity the concordance

remains 1 down to a linear tail of about 17% of rows, whether the change is abrupt or spread over four batches. At a tail of about 8%, full abrupt intensity falls to mean concordance 0.850 with a worst seed of 0.750, so the mean still clears 0.8 and the guarantee across seeds does not. Gradual activation at that thin tail has the same character (0.867, worst seed 0.800). Lowering the rewrite probability to 0.7 preserves the target on the main tail and on the 17% tail. Combining $\alpha = 0.7$ with the 8% tail and a four-batch ramp drops the concordance to 0.683, with a worst seed of 0.500. At $\alpha = 0.4$ every column is below 0.8. The main-tail abrupt entry is 0.783 with a worst seed of 0.650. On that same $\alpha = 0.4$ slice, the whole-window score without the tail scan has concordance 0.433. The region maximum is what keeps a weak but structured rewrite near the target. It does not keep an arbitrarily weak rewrite above the target.

2.8 What the benchmark supports

The benchmark supports four claims, each tied to a table. First, when the rewrite is complete and the linear tail contains about a fifth of the rows or more, Δ^{cf} orders every structured site ahead of every clean site and ahead of the noise site, on two real covariate tables, in every seed (Table 19). Second, the functionals that ignore either the existence of a local concept or the region in which that concept lives do not. Label total variation and the covariate arms are near chance against clean sites. The joint two-sample test and the pooled Brier excess promote the noise site, completely so for linear sites under the pooled excess on Income. Third, the time profile of the same score locates an abrupt change to within about two batches once the scoring window is taken into account, with a false-alarm rate on clean Income sites that a deployment threshold would still have to budget for (Table 20). Fourth, a pre-specified concept-group dictionary can name the nonlinear mechanisms from the full window and the linear mechanism from the tail, and it cannot do both from a single region (Table 21).

The benchmark does not support a claim that leave-one-site-out influence on an averaged variable-importance vector identifies the primary site. Those vectors are too similar across sites for the deletion of one upload to move the rank correlation. It also does not support a claim at very small intensity. Below a rewrite probability of one half, or inside a linear tail that covers less than a tenth of the window, pairwise concordance falls through 0.8 on at least one seed. Both limits are visible in the same design that produces the perfect separation at full intensity, which is the reason for reporting them beside the headline rather than as a separate simulation.